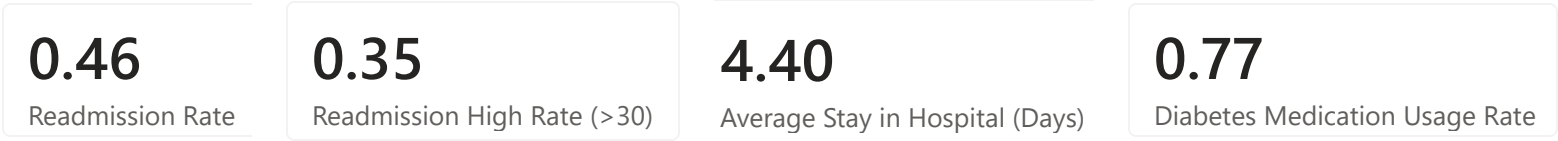


Hospital Readmission Risk Analysis and Prediction

(Diabetes 130 US Hospital Dataset)



Age

All

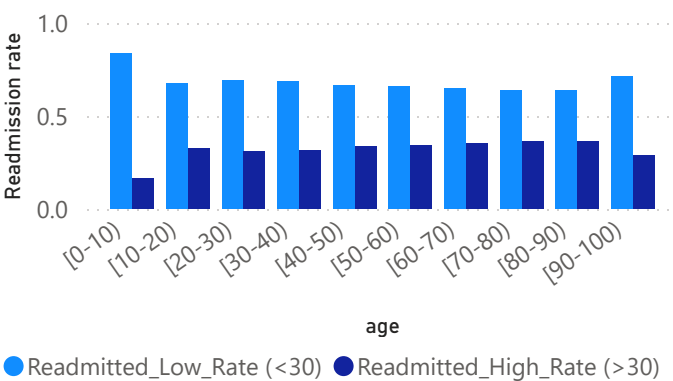
Readmission Status

All

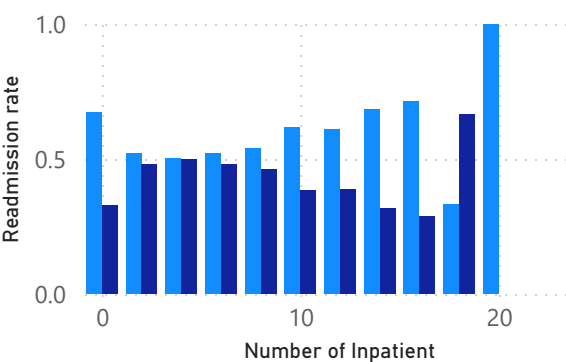
Diabetes Med Stat...

All

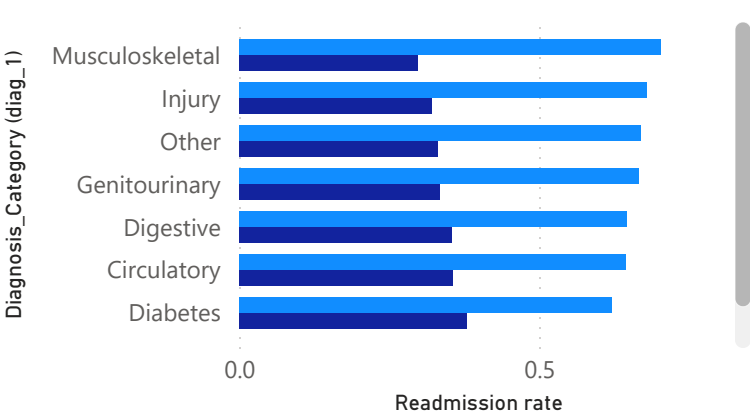
Readmission Rate by Age Group



Readmission Rate by Number of Inpatient

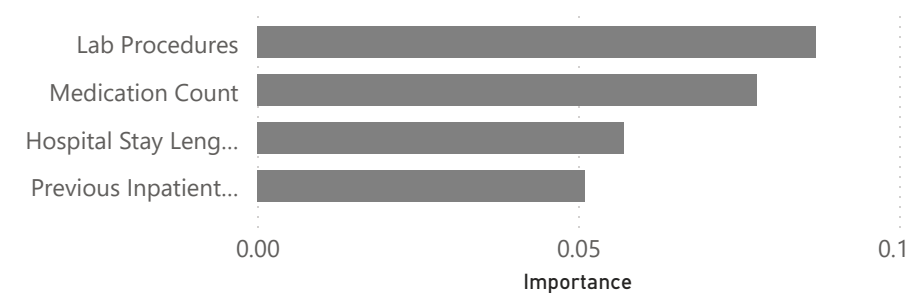


Readmission Rates by Diagnosis Category (diag_1)

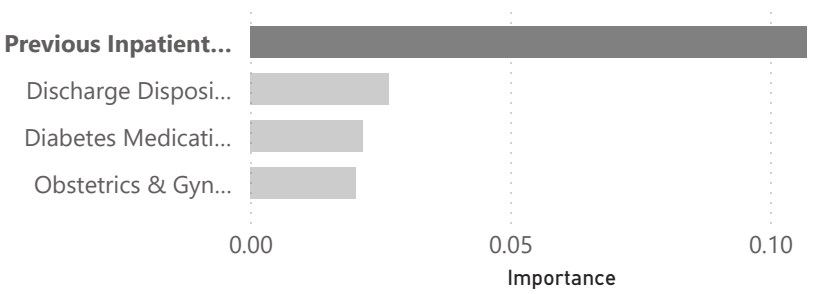


Top Features Influencing Readmission Risk Using Machine Learning Models

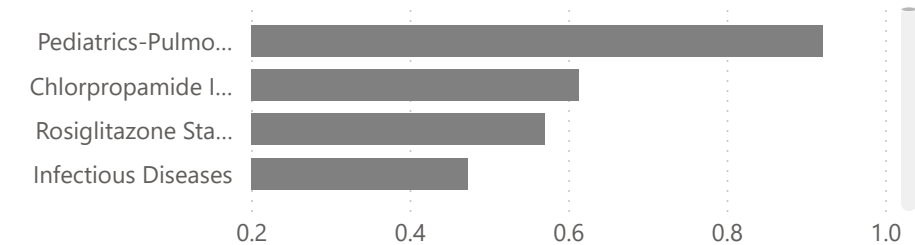
Random Forest



XGBoost



Logistic Regression



Model	Accuracy	Precision	Recall	F1	ROC_AUC
Logistic Regression	0.62	0.63	0.44	0.52	0.66
Random Forest	0.63	0.62	0.52	0.56	0.68
XGBoost	0.65	0.63	0.56	0.59	0.70

> **XGBoost** achieved the strongest predictive performance (ROC-AUC = 0.70), outperforming Logistic Regression and Random Forest in identifying patients at risk of hospital readmission.

Summary

- 1) Patients aged **70–90 years** exhibited the highest readmission risk.
- 2) **Higher readmission risk** was associated with longer hospital stays, greater medication counts, and a higher number of diagnoses.
> These findings suggest that patients with more complex clinical conditions may require enhanced monitoring and follow-up care.
- 3) **Random Forest** identified previous inpatient visits, laboratory procedures, medication count, and hospital stay length as key predictors of readmission risk.